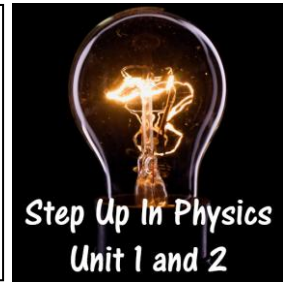


Heat Transfer Mechanisms

Problems Worksheet



1. When stirring a sauce that has been heated on a stove, Jane notices that the end of her metal spoon gets hot quickly but the end of a wooden spoon takes a long time to get hot.
 - a. Why do the ends of the spoons increase in temperature?

 - b. Why does the metal spoon get hot faster than the wooden spoon?

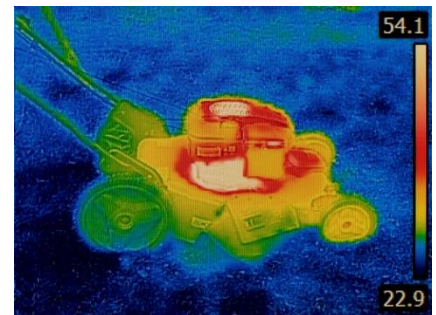
2. When someone is rescued after spending a long time in the ocean the paramedics will wrap the person in a thermal blanket; a silver coloured sheet.
 - a. What is the purpose of the thermal blanket?

 - b. Which method of heat transfer is affected by a silver coloured surface?

- c. The sheet is silvered coloured on both sides. Which side (facing person or facing air) is most important that it has a silver covering? Explain why.
 - d. Jumpers and jackets do not need to be silver coloured to keep a person warm. Explain how a jumper assists in keeping a person warm.
3. Most homes in Australia are built with double glazed windows. These are two panes of glass stuck together with an air gap between them. Explain why double glazed windows are useful in both summer and winter months.

4. Hydrothermal ocean currents are formed around areas of underwater volcanic activity. This is where the water flows vertically. Explain how these currents are formed.

5. The image below is a thermal image. It has captured information about the temperature of the objects in its field of view. How is a thermal camera able to measure information about the temperature of the lawn mower even when at a distance?



6. When waking up on a cold morning, Jane stands on her carpeted bedroom floor, hesitant to stand on the tiled bathroom floor. Both the carpet and tiles are at the same temperature yet the tiles feel much colder than the carpet. Explain.

7. Many homes in Australia use solar hot water systems placed on the roof. These systems are carefully designed to maximise the amount of solar energy that is transferred into the water.



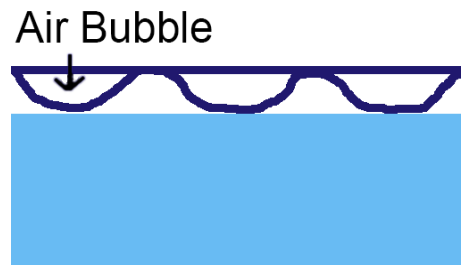
Water travels up tubes in the flat, dark coloured panel before being stored in the light coloured storage tank.

- a. Explain the purpose of the **dark** panel of a solar hot water system.

- b. Explain the purpose of the **light** storage tank of a solar hot water system.

8. Hang gliders use rising pockets of air (called thermals) for gaining altitude even though they do not have a powered thrust source like planes do. Explain how thermals form.

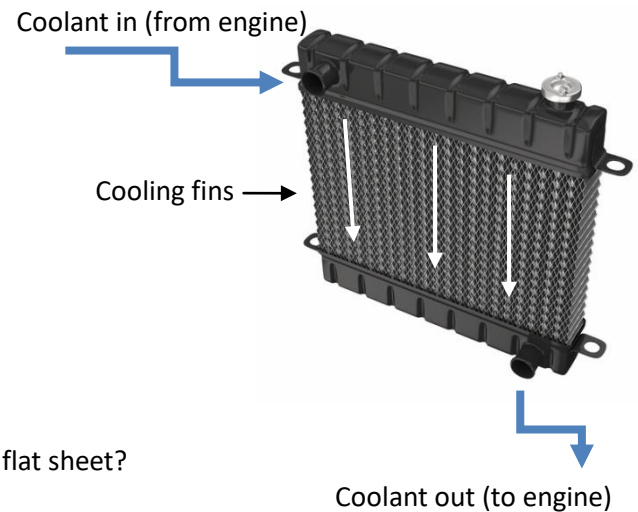
9. A pool blanket is designed to make a pool feel more comfortable by increasing the water's temperature a few degrees compared to a pool with no blanket. The design of the pool blanket is shown below.



- a. The pool blanket is a passive device; it requires no power to perform its function. Where does the energy to heat the pool originate? How does this energy enter the pool?
- b. Describe one means by which the pool blanket limits heat loss from the water.
- c. When the blanket is removed, swimmers often claim the surface is very warm while the deeper water is still cool. Suggest a reason for this.

10. A car's radiator system is important as it keeps the engine cool enough to keep running. The excess heat of the engine is passed onto the coolant. The heat is then released from the coolant to the atmosphere at the cooling fins.

a. Why are the cooling fins a dark colour?



b. Why is a rough surface of fins used and not a flat sheet?

11. A thermal flask is capable of keeping hot contents hot by minimising heat loss by conduction, convection and radiation. Explain how the thermal flask minimises heat loss for each method of heat transfer.

